



PROCESSOR

From wasted clock cycles to inefficient programs, there are many ways your processor can slow down your PC

The veritable brain behind your computer, the processor is the first component associated with the speed of a computer. The common fallacy is that the speed of a processor (and eventually the computer) is solely dependant on its megahertz (MHz) rating—the higher the rating, the faster the various sets of data and instructions are processed. Clock speed does play an important role in determining the processor's speed, but it is not the only factor that affects its performance. There are several other subcomponents in a processor that can spell the difference between your computer taking 10 seconds more to boot up, or your 3D model taking 1 minute more to render in 3D Studio Max.

The bus speed supported by your processor is also an important determinant of your processor's speed. For example, when the Pentium 4 processor was first launched, it was based upon a 400 MHz system bus. However, in its newest iteration, the Pentium 4 uses a 533 MHz system bus. This means that data would move between system components much faster and you would see the benefits in applications such as Adobe Photoshop, which make heavy use of the three most important subsystems—the processor, the RAM and the hard disk. Therefore, if you upgrade to a processor that supports higher bus speeds, you would benefit not only from the increase in the processor's frequency, but also from the faster data transfer rates between the computer's subsystems.

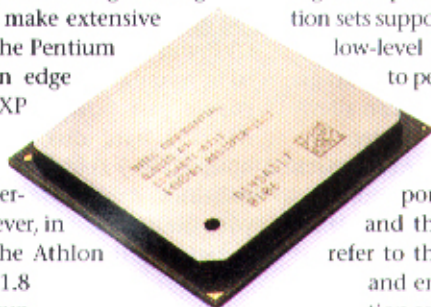
Ever wondered why even though you have roughly the same system configuration as your friend, the applications seem to run faster on his computer, the only difference being that he has a different brand of processor?

One reason is the amount of cache the processor carries.

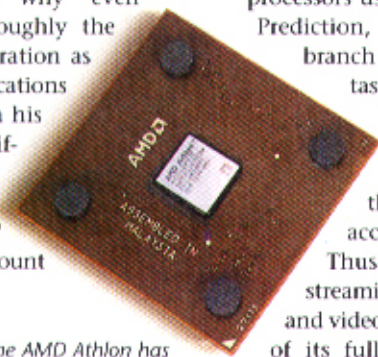
The two types of cache are Level 1 (L1) and Level 2 (L2). Cache is defined by its size and the speed it is running at. For example, in games like *Quake III* and image editing applications that make extensive use of L2 cache, the Pentium 4 would have an edge over the Athlon XP due to the greater amount of cache (512 KB versus 256 KB). However, in MP3 encoding, the Athlon XP running at 1.8 GHz would outrun even a 2.4 GHz Pentium 4! This is due to another processor subsystem, the integer and floating point units. It is here that all the actual calculations take place and with the Athlon XP having three floating point and integer units each, compared to the Pentium 4 which has only one of each, the Athlon XP would definitely come out tops.

Among the Intel and AMD clans of processors, two perceptible routes of processor design have emerged—Intel designs its processors with longer pipelines (a section of the processor that is fed with a flow of instructions) to allow for higher clock speeds; AMD still uses shorter pipelines, which means it will not be able to touch clock speeds as high as Intel. In a practical situation, when both the processors need to choose which branch to take while executing a task, the length of the pipelines comes into play. All newer processors use a technique called Branch Prediction, where they guess which branch would be most suitable for a task, before the actual decision point. In the event that a wrong choice is made, the Intel processor would face the biggest performance hit on account of its longer pipelines. Thus, the Pentium 4 is ideal for streaming applications such as audio and video authoring, but will fall short of its full potential performance in non-streaming software such as office applications.

We often come across terms such as MMX, SSE2, 3D Now!, SIMD, etc when talking about processors. These are the instruction sets supported by the processor, i.e. the low-level language the processor uses to perform tasks such as addition, multiplication, data movement, etc. Different brands of processors support different instruction sets and the terms mentioned above refer to the various implementations and enhancements to the instruction sets of popular processors from Intel and AMD. Sadly however, these hi-tech sounding acronyms have meant precious little in real world terms. For example, when MMX-enabled processors were launched, people thought that all their applications would magically run faster. In reality, unless the applications you use specifically state support for these instruction sets, you will not perceive any difference in their speed of operation.



Intel's Pentium 4 processor boasts of higher clock speeds



The AMD Athlon has six number-crunching pipelines

Tips & Tweaks

- Verify that your processor's L2 cache is enabled in your BIOS.
- Load specific patches for your applications, which tap into special instruction sets on your computer. You can find them on the respective application sites such as Adobe and Discreet, for example, where they have specific support for newer processor technologies with each new iteration of their application software.
- See that your processor has adequate cooling in terms of the CPU fan and system airflow. Use a large heatsink with a good quality CPU fan and ensure that the airflow in your system cabinet is optimal. This prevents lockups and application crashes caused by excessive heat generation. Older processors such as those based on 0.18-micron fabrication processes generally use higher core voltages and therefore tend to generate more heat. Special care needs to be taken with these processors.